

Mudda: AI Social Media Governance Platform



Mudda

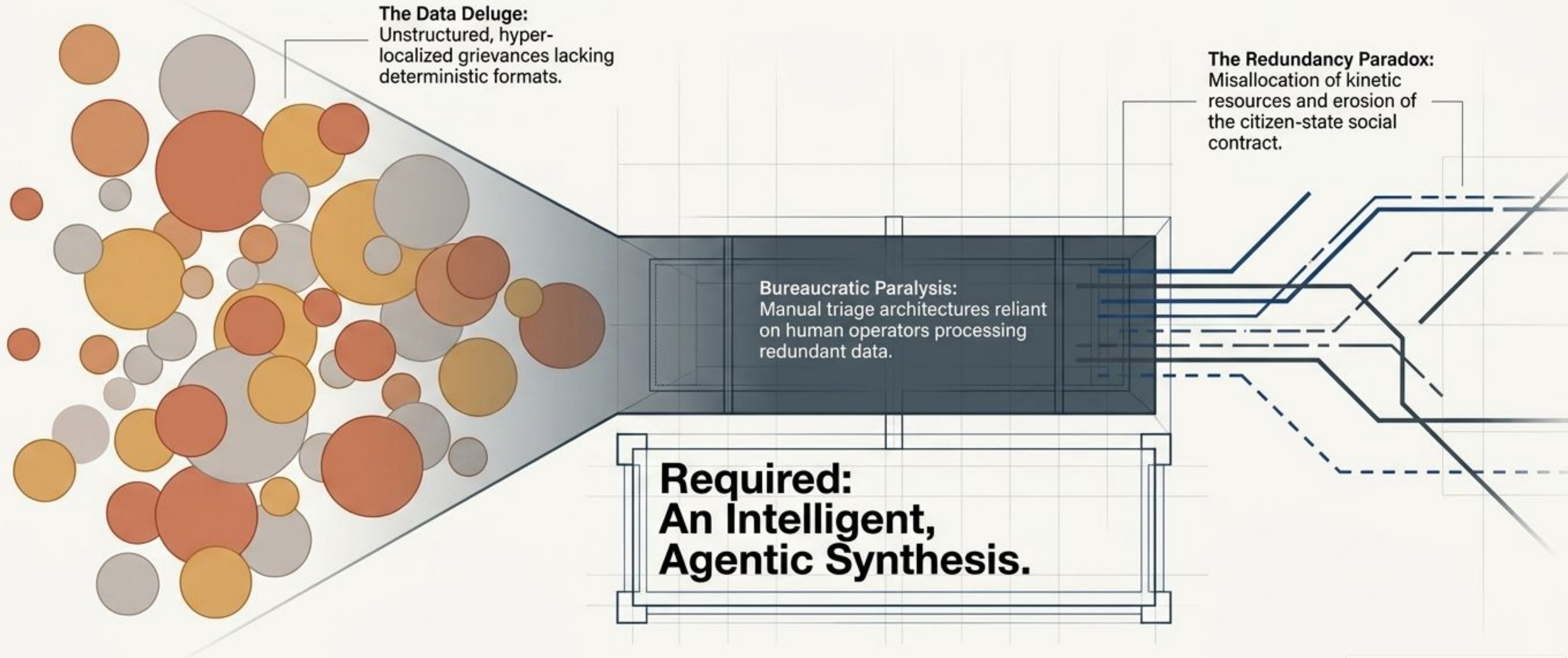
Supervisor (s):
Dr. Shilpa Mahajan

- Team Member(s):
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Shivesh Kumar Dhiman (22CSU164)
Shubh Pundir (22CSU165)
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Department of Computer Science and IT

Problem Statement

The epistemological breakdown between citizens and the state creates operational paralysis.



Unstructured Social Talk

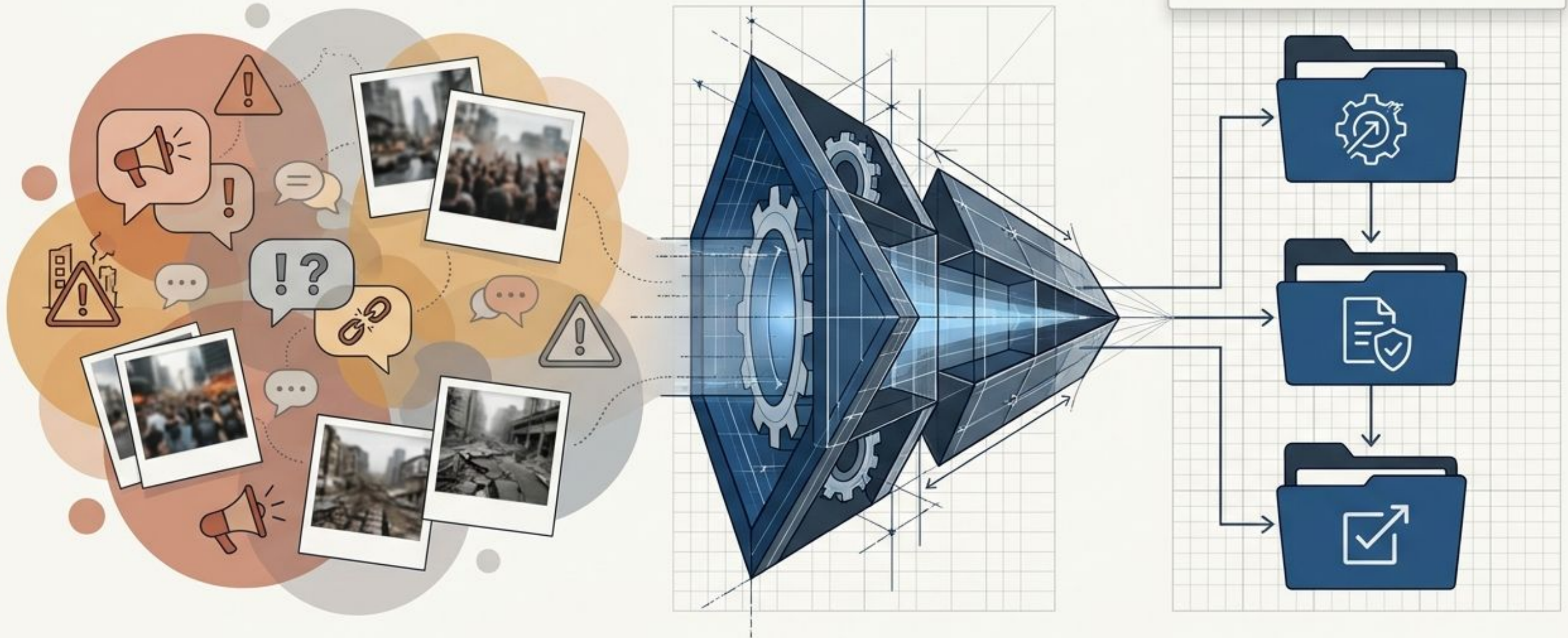
Multi-modal inputs: Tweets, photos, informal complaints.

Mudda AI Operating System

Spatial-semantic clustering and sentiment assessment.

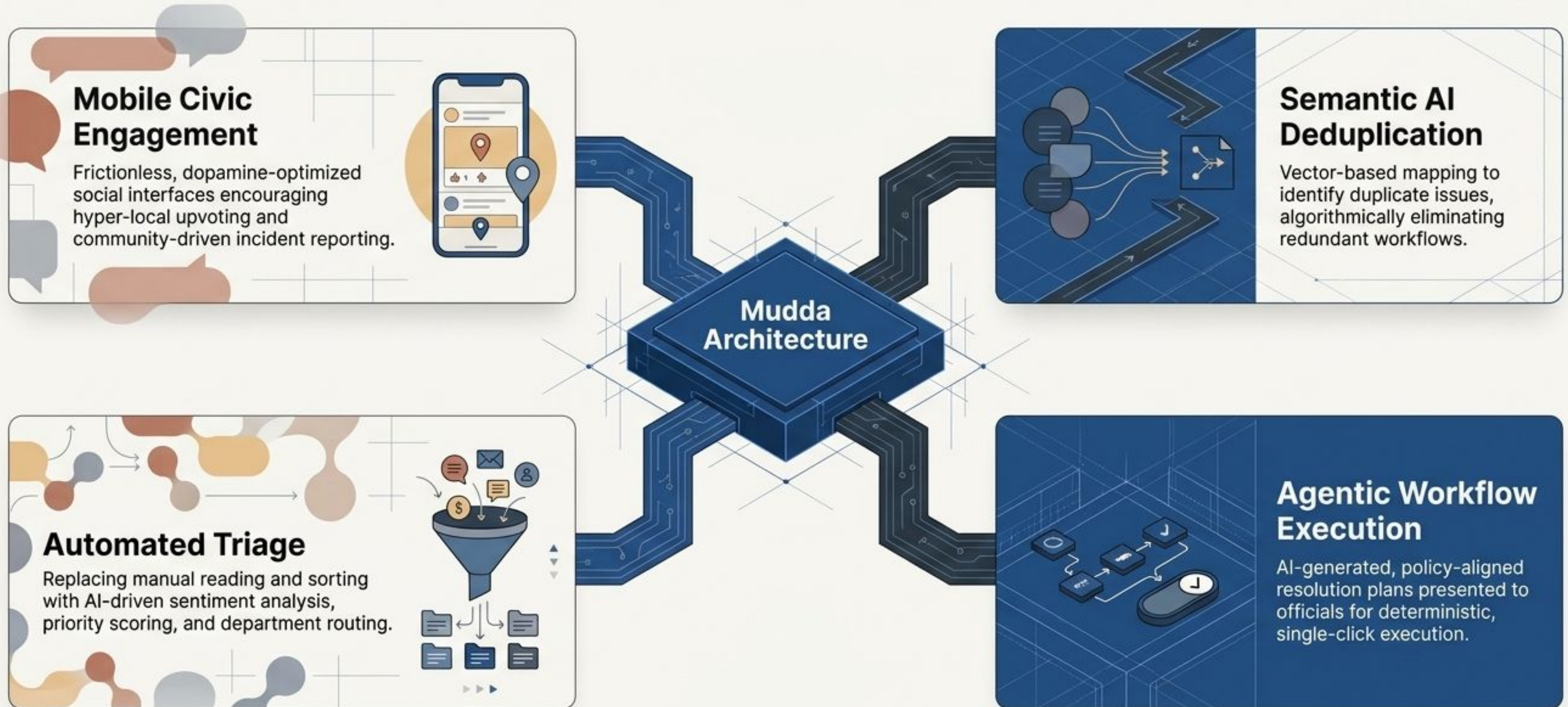
Structured Administrative Workflows

Agentic issue routing, policy alignment, and executed resolutions.



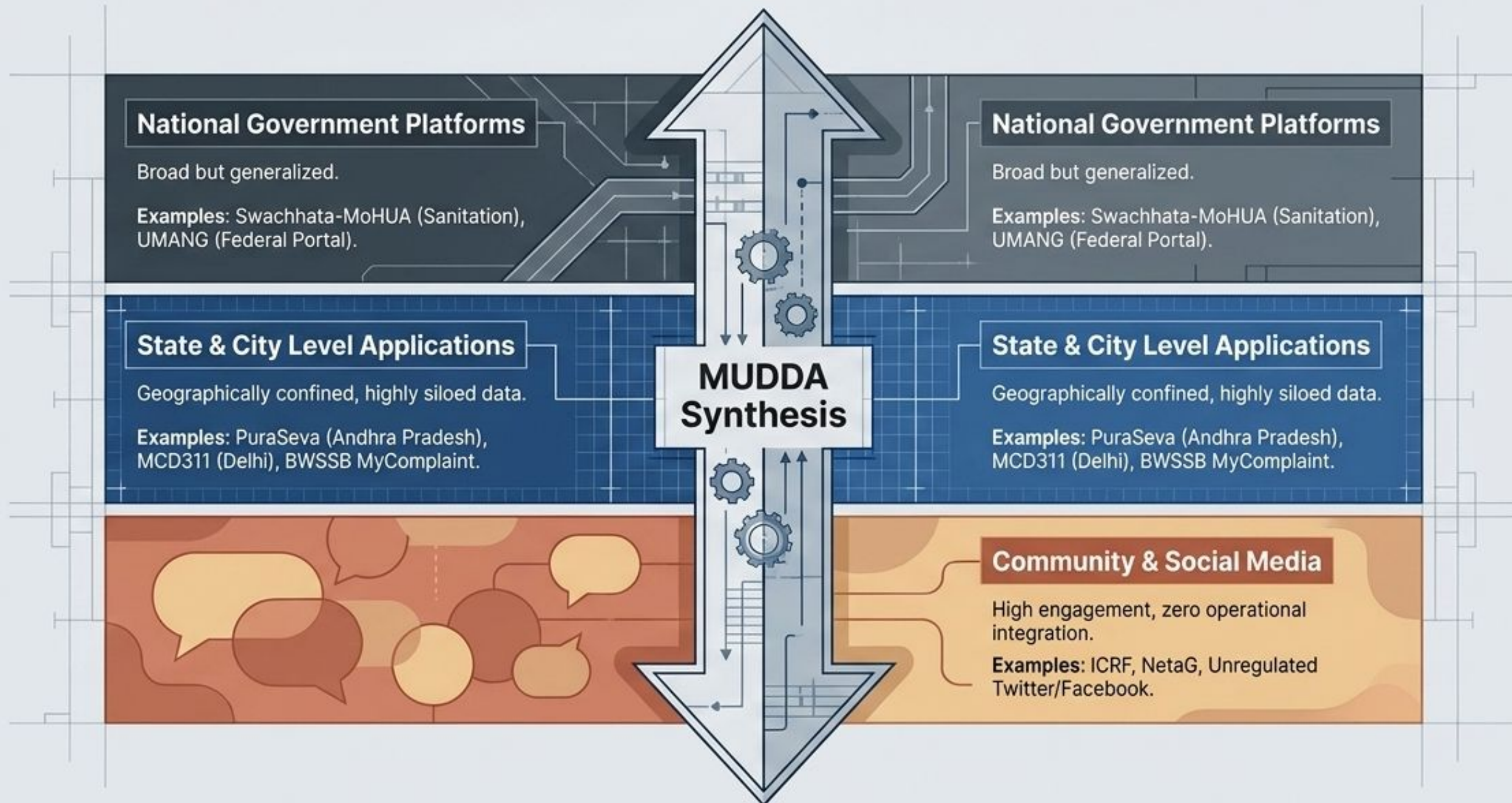
Core Objectives

Translating the collective voice of the masses into optimized civic execution.



Landscape of Existing Solutions

Reviewing current civic grievance mechanisms across three distinct tiers.



Gap Analysis

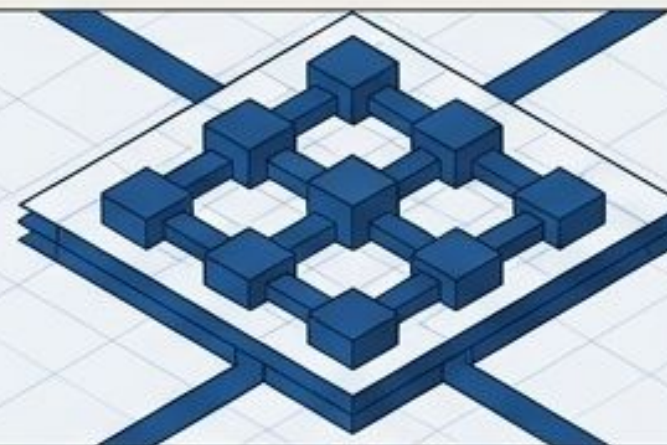
Major inefficiencies in legacy systems

Diagnostic Matrix

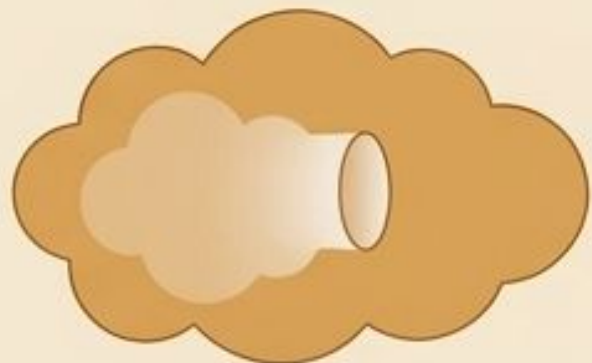


The Intelligence Gap

Legacy systems lack semantic spatial awareness, leading to multiple crews dispatched to a single localized failure.

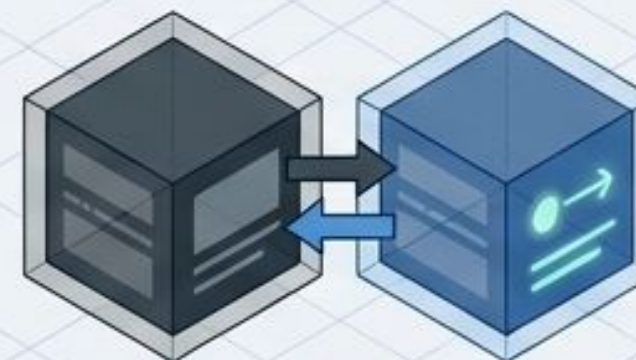


Mudda uses spatial-semantic clustering to mathematically eliminate the Redundancy Paradox.

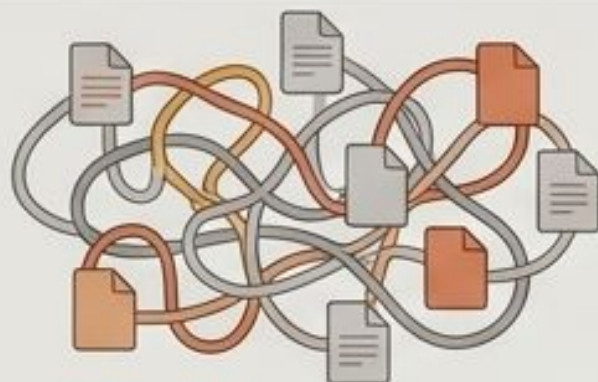


The Communication Gap

Current apps act as opaque data receptacles, providing asynchronous and vague status updates.



Mudda provides bi-directional, synchronous updates driven by optimistic UI states.



The Orchestration Gap

Existing tools demand entirely manual dispatch drafting and manual policy referencing.



Mudda introduces Agentic Assistive AI workflows for proactive, single-click authorization.

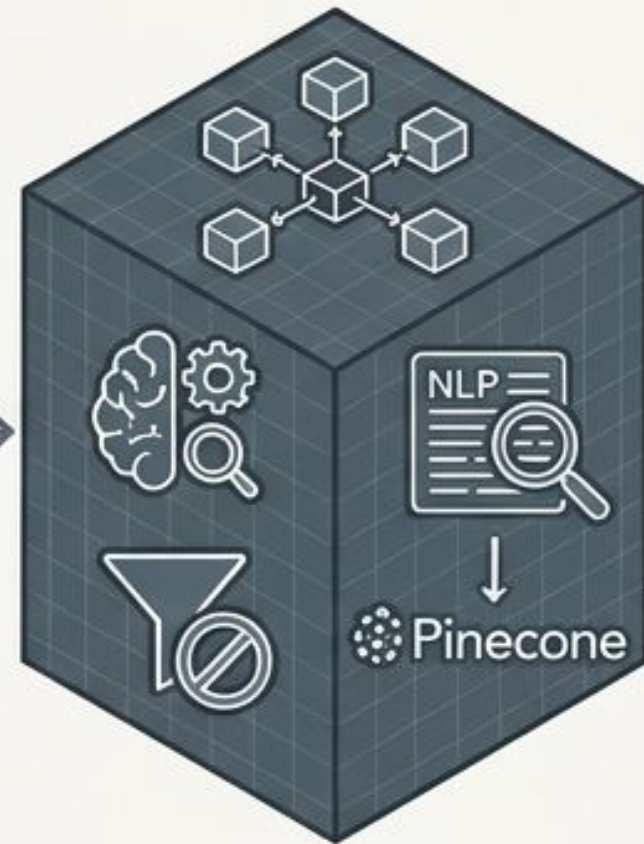
Architectural Blueprint

The AI-Powered Civic Engagement Pipeline.



Phase 1: Edge Data Ingestion

Mobile Flutter UI.
Unstructured multi-modal
inputs, geotags, images, text.



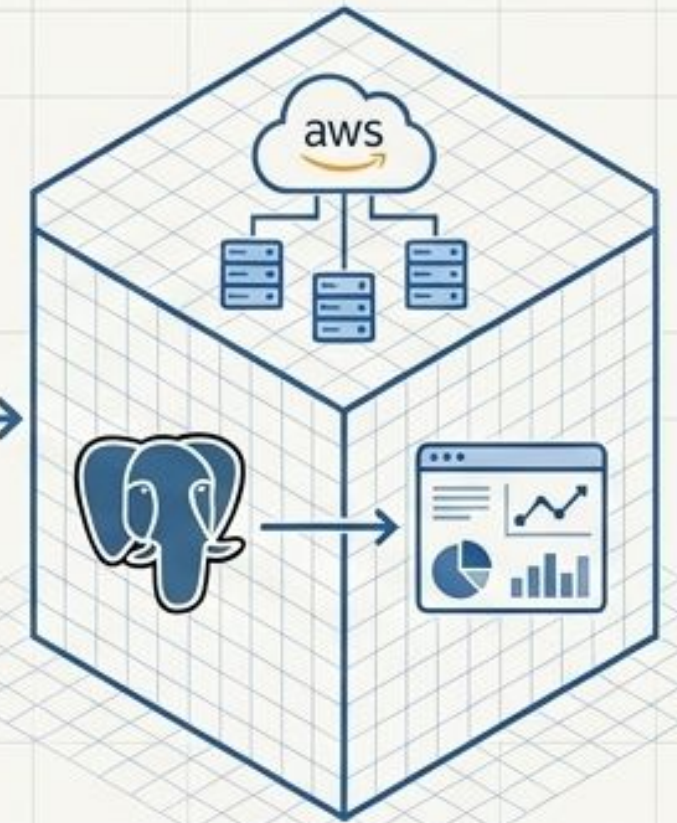
Phase 2: Core AI Brain

Python microservices.
NLP metadata extraction,
hate-speech filtering, and
Pinecone vector deduplication.



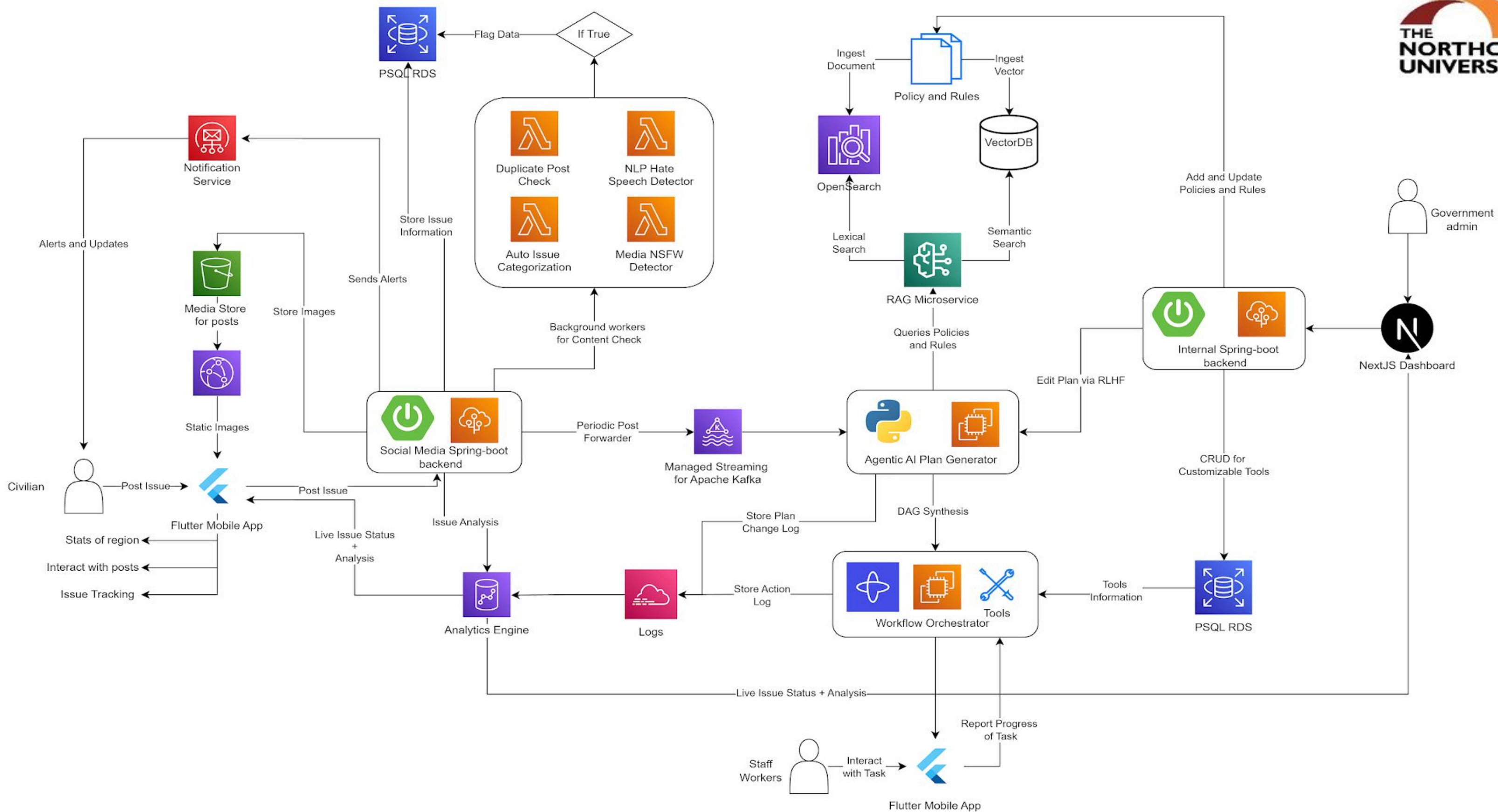
Phase 3: Temporal Orchestration

**Asynchronous event-driven
mapping.**
Converting probabilistic AI
outputs into deterministic
execution plans.



Phase 4: Output & Visualization

AWS infrastructure.
PostgreSQL commits pushing
zero-latency updates to Next.js
municipal dashboards.



Design Methodology

Iterative, feature-first development via concurrent workstreams.

Frontend UI Engineering

Iterative mobile screens and web layouts. Optimistic UI updates to mask backend processing latency.



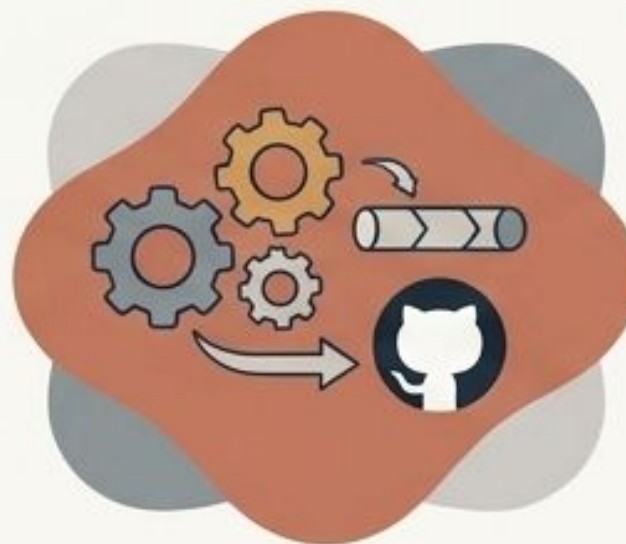
Microservices & Cloud

Spring Boot API development, AWS scaling, and PostgreSQL schema locking.



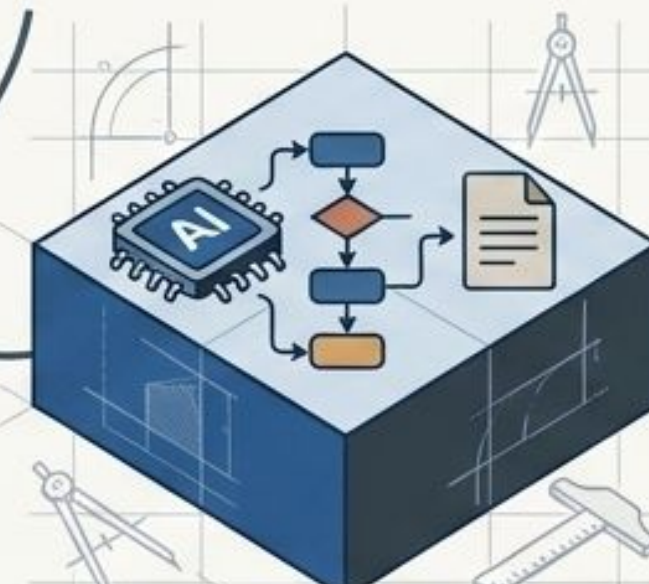
DevOps & Deployment

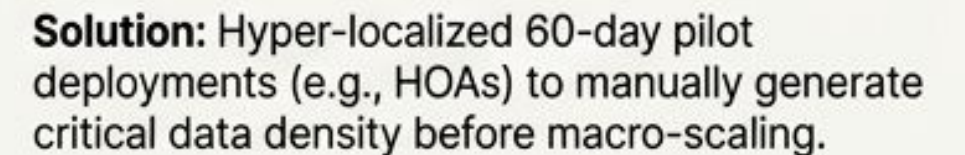
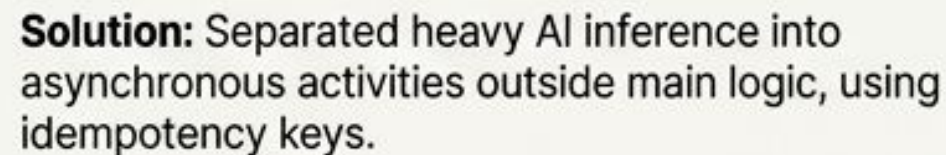
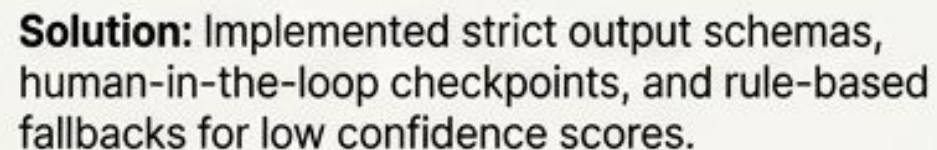
GitHub Actions CI/CD pipelines ensuring seamless integration and zero-downtime updates.



AI & LLM Orchestration

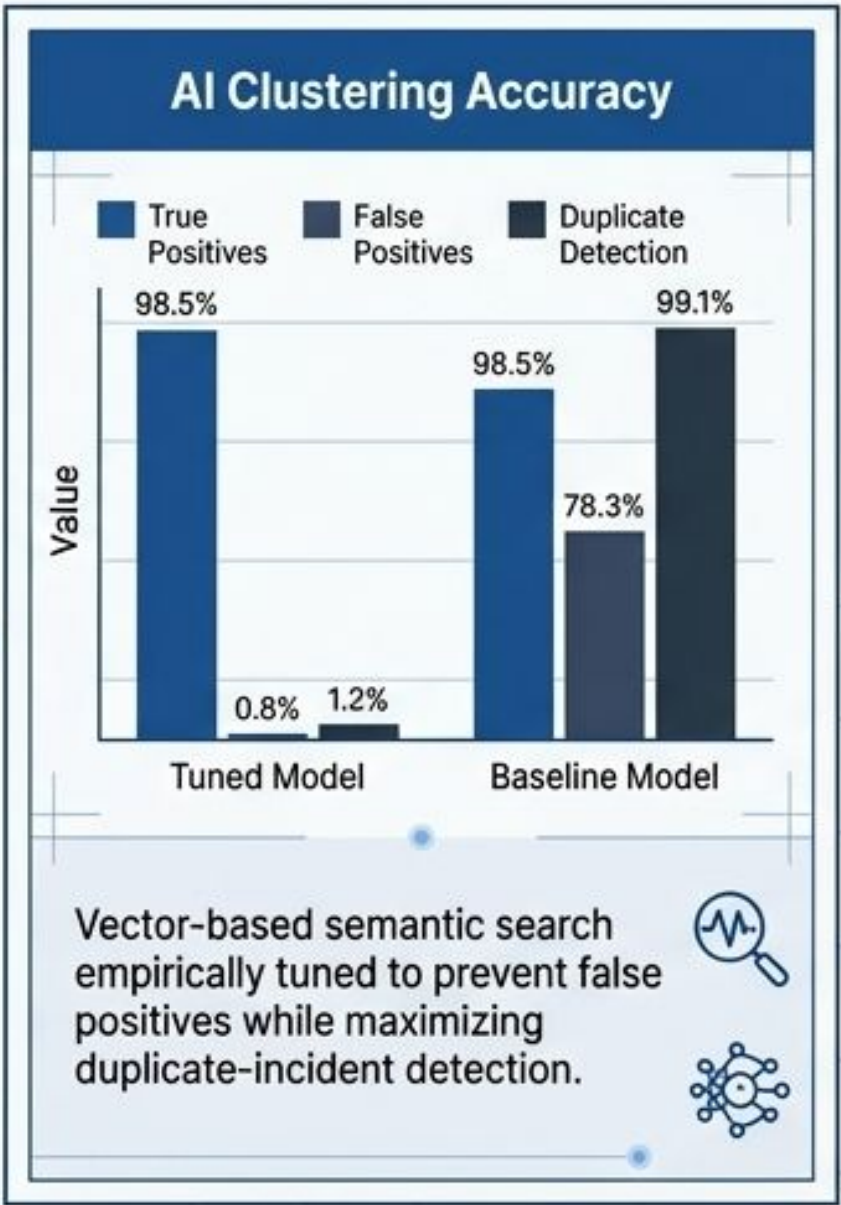
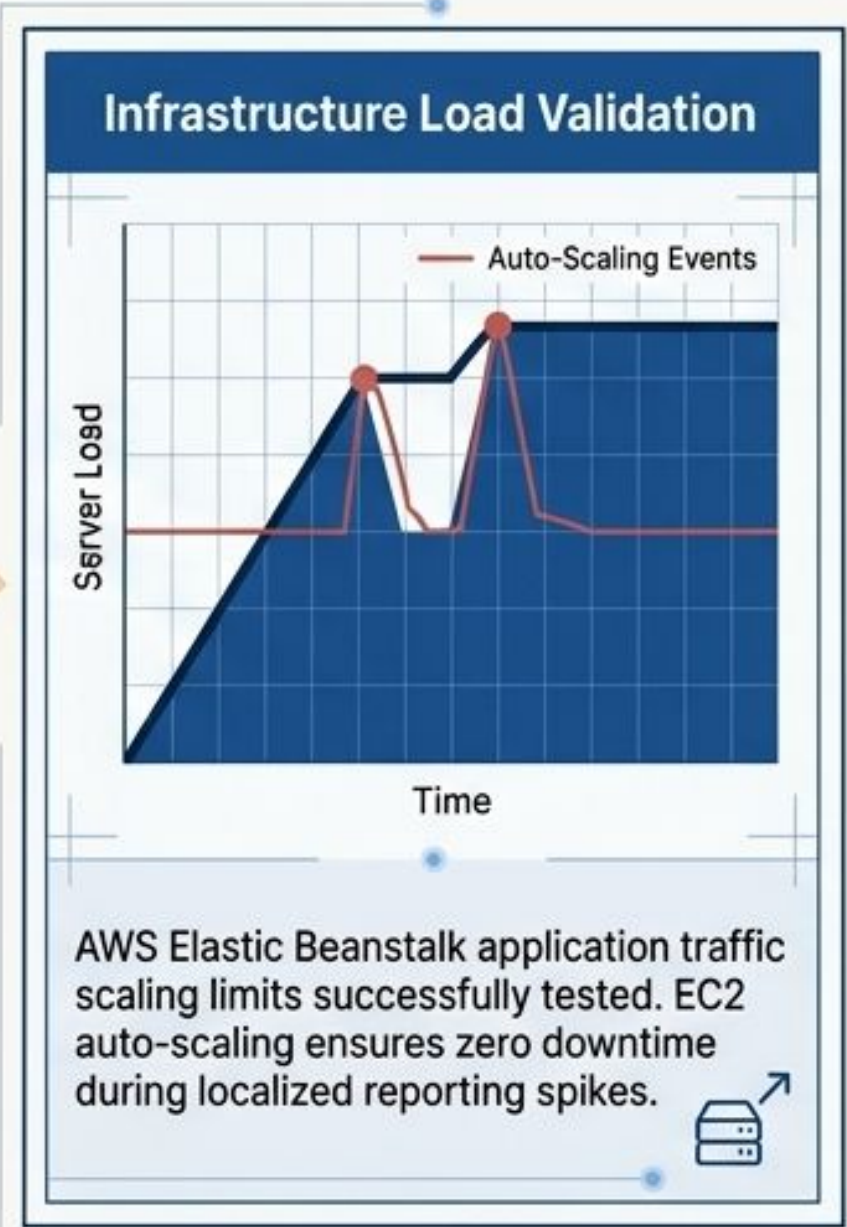
Asynchronous event-driven mapping of AI outputs into deterministic, policy-compliant execution plans.





Testing and Performance Evaluation

Validating the system through rigorous empirical benchmarks.



Citizen UI/UX Acceptability

Feature Tested	UAT Result	Status
Infinite-Scroll Feed	Confirmed	Approved
Offline-First Resilience	Local Caching Active	Approved
API Pagination	Confirmed	Approved

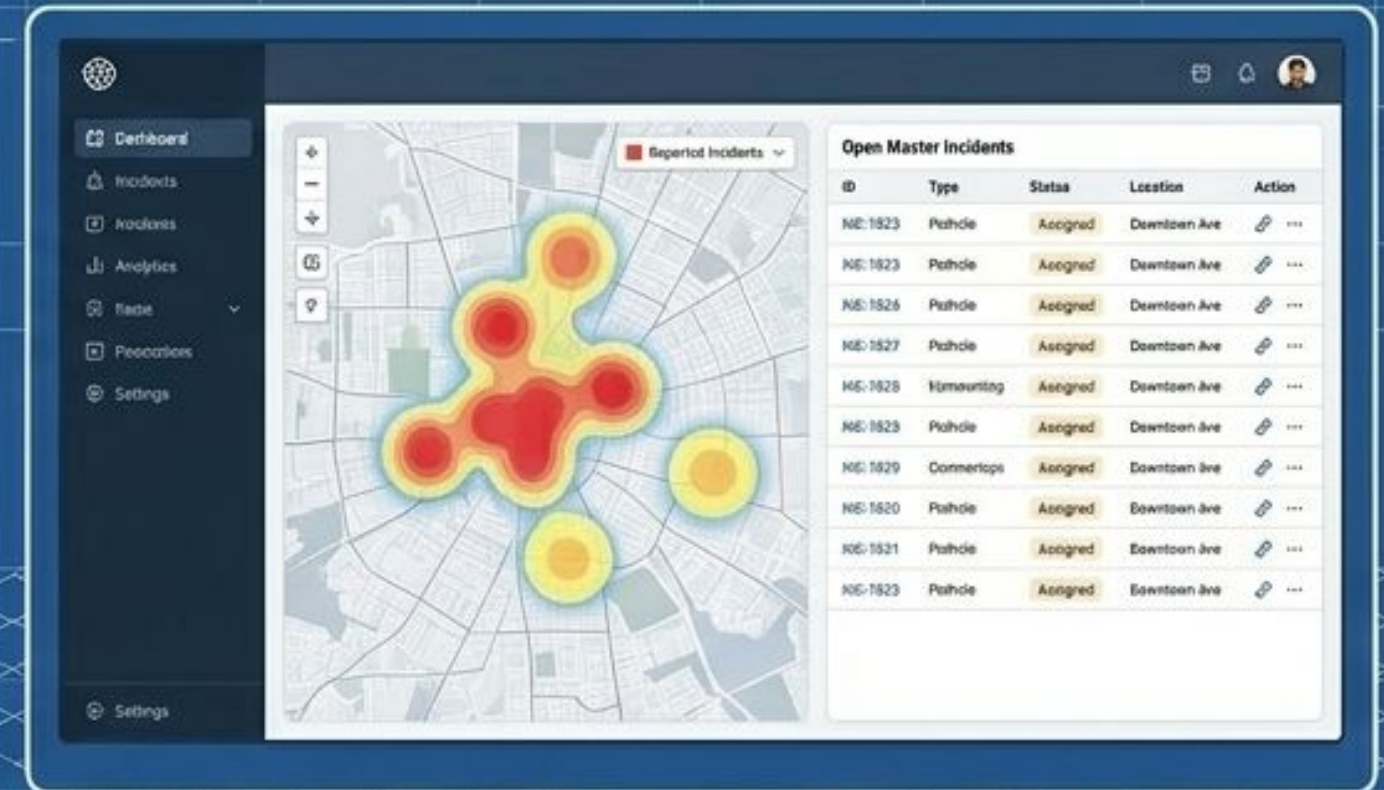
UAT confirms infinite-scroll feed memory management (API pagination) and offline-first resilience via local caching.

Dual-Engine Interfaces

The Citizen Experience vs. The Government Experience.



Frictionless, dopamine-optimized civic engagement powered by optimistic UI updates.



Deterministic web dashboard for agentic, policy-compliant AI workflow orchestration.

Expected Outcomes

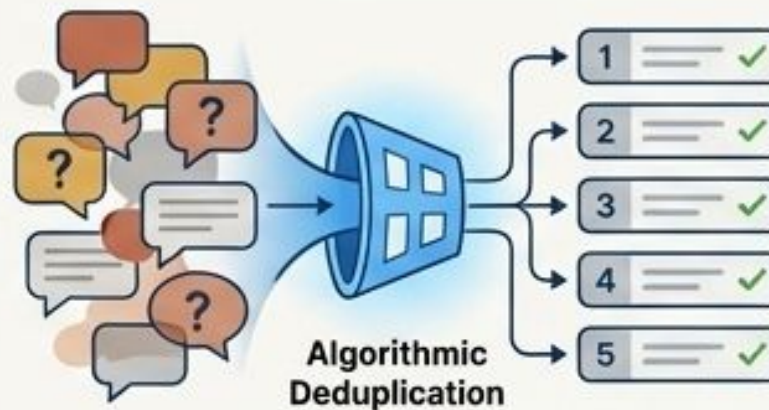
Delivery of a pilot-ready, socio-technical shift.

1 The Prototype



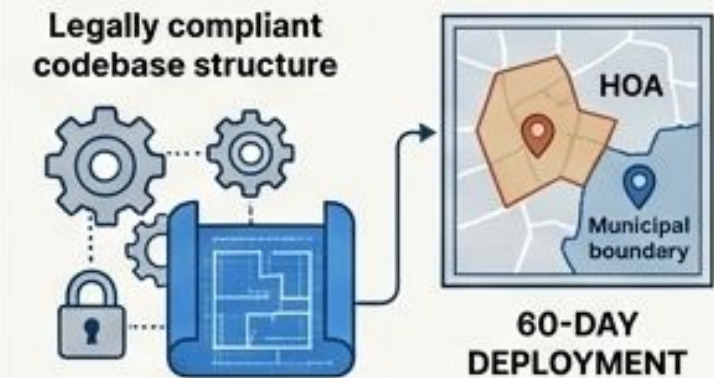
A fully functional, dual-engine prototype merging an interactive Flutter mobile frontend with an authoritative Spring Boot backend.

2 Administrative Relief



Algorithmic deduplication minimizes manual triage, transforming chaotic public sentiment into a streamlined, automated operational reality.

3 Pilot Readiness



A legally compliant, scalable codebase architected specifically for targeted 60-day hyper-localized deployments in HOAs and municipal zones.

Feasibility Study

Establishing the viability of assistive AI for duplicate complaint clustering

Technical Feasibility

Single-codebase cross-platform parity via Flutter. Auto-scaling computational intelligence via AWS and Spring Boot architecture.

Operational Viability

Assistive AI cognitively offloads manual triage without usurping municipal decision-making. Frictionless API integration with legacy systems.

Economic Viability

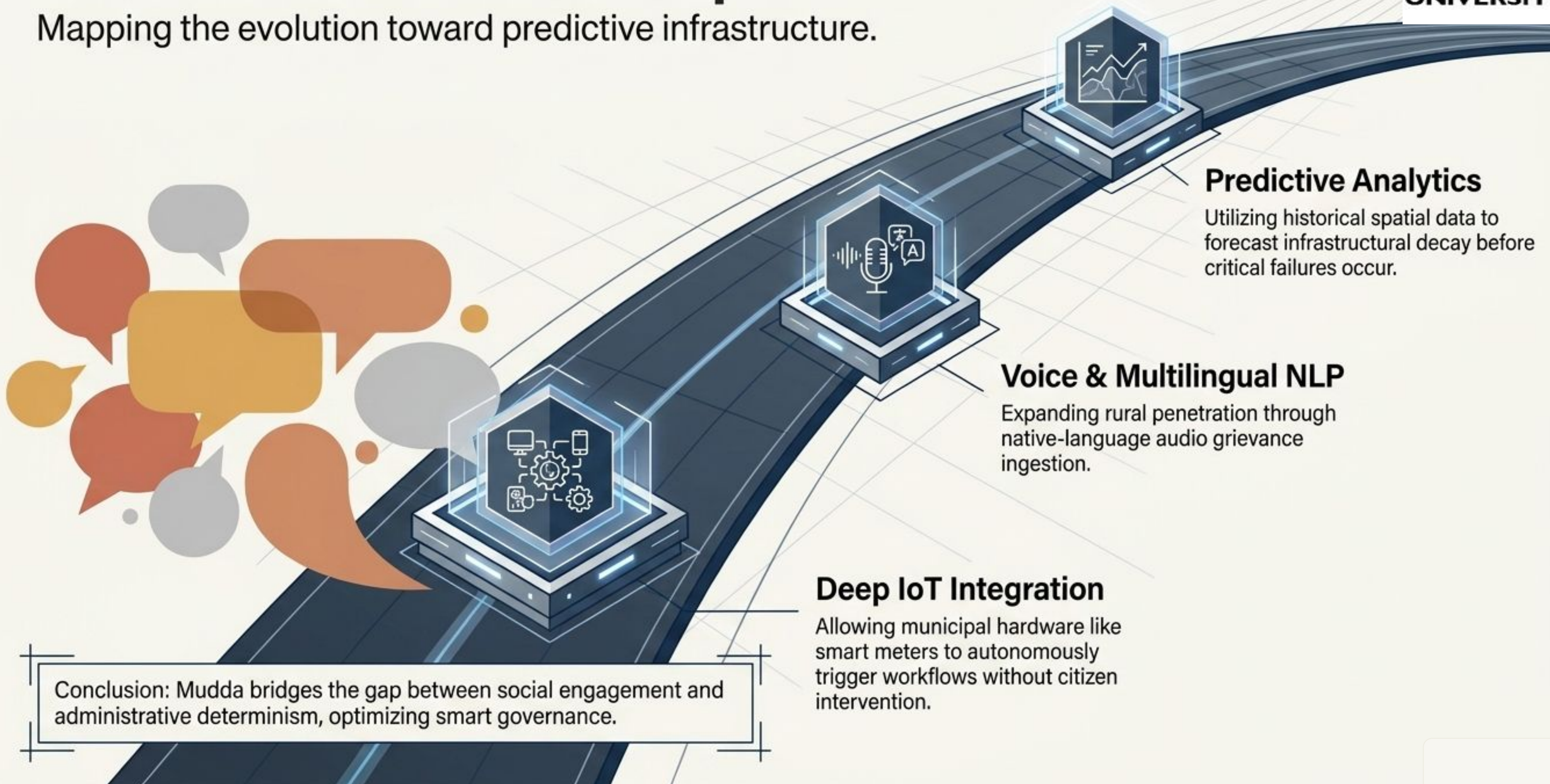
Mathematical elimination of redundant kinetic resources. Highly scalable B2G SaaS monetization targeting high-density municipalities.

Legal & Ethical Viability

Strict adherence to DPDP Act, 2023. Personally Identifiable Information is completely obfuscated before spatial-semantic AI processing.

Conclusion & Future Scope

Mapping the evolution toward predictive infrastructure.



Technology Stack & Unique Selling Propositions

Frontend

NEXT.js Flutter



Stack: NextJS (mudda.dev), Flutter (mobile app)
USP: Web dashboard for admin, mobile app for civilians and ground staff workers.

Backend

python spring



Stack: Python, Spring
USP: Scalable microservices for CRM maintaining and traceability.

Database & Search

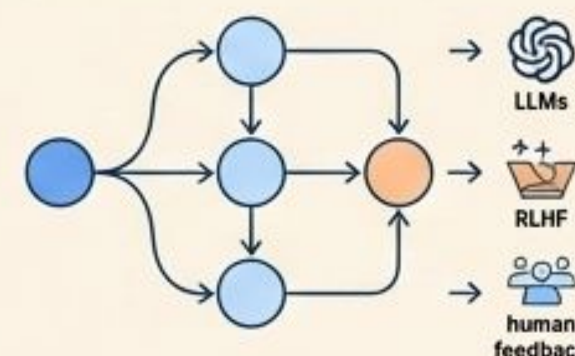
PostgreSQL Qdrant (VectorDB) Elastic Search



Stack: PSQL, Qdrant, Elastic Search
USP: Queries out important and exact references via hybrid search for less hallucination, refers actual pdf docs with highlighted lines for policies, past context, past cases understanding, etc.

AI & Orchestration

Langchain Langsmith Temporal



Stack: Langchain, Langsmith, Temporal
USP: Traceability, Predictability -> LLM orchestration is different from system orchestration, RLHF, Human in the loop, memory condensation after each step of temporal execution.

Broader Impact Validation

Quantifying the real-world value of Mudda's deployment.

Resource Optimization

Before:
Ordundant vehicle
and crews.



Before:
Redundant vehicle
and crew dispatches
to single locations.



After:
Mathematical
elimination of overlap
via AI spatial-semantic
clustering.

Resolution Acceleration

Before:
Extreme latency due to
manual reading and
heuristic triage.



Before:
Extreme latency due
to manual reading and
heuristic triage.



After:
Drastic reduction in
SLA response times
through automated
pre-filtering.

User Acceptance

Before:
Civic apathy and
opaque black-box
portals.



Before:
Civic apathy and
opaque black-box
portals.



After:
High community
engagement driven by
real-time tracking and
optimistic UI responses.

References & Benchmarks

Regional municipal portals and national public platforms evaluated.

Federal & State (Governmental Structures)

Swachhata-MoHUA

Focuses exclusively on sanitation; lacks comprehensive sectoral routing.

UMANG

Federal umbrella portal; not optimized for hyper-local spatial grievance tracking.

PuraSeva (Andhra Pradesh)

State-level solution but suffers from siloed deployment.

City & Community (Localized & Unregulated Platforms)

MCD311 (Delhi)

City-limited, traditional legacy architecture relying on manual dispatch.

ICRF & NetaG

High community engagement but zero integration with actual municipal resolution pipelines.

Responsibility Matrix

A specialized engineering team balancing UI, DevOps, and Orchestration.

Saumil Kulshreshtha

Frontend Lead

Flutter UI/UX architecture, media payload integration, and API mapping.



Vikas Kumar

Backend Lead

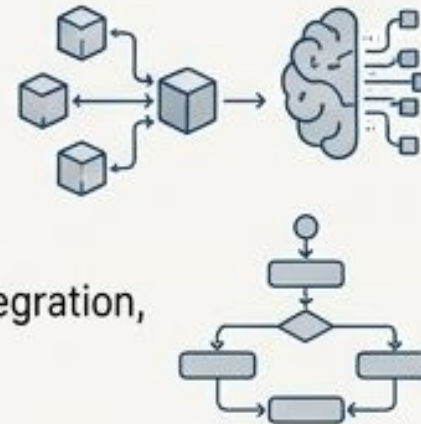
Spring Boot schemas, REST API security, and database relational integrity.



Shubh Pundir

AI Models / Backend

Python microservices, LLM orchestration, Temporal integration, semantic deduplication.



Shivesh Kumar Dhiman

DevOps / Frontend

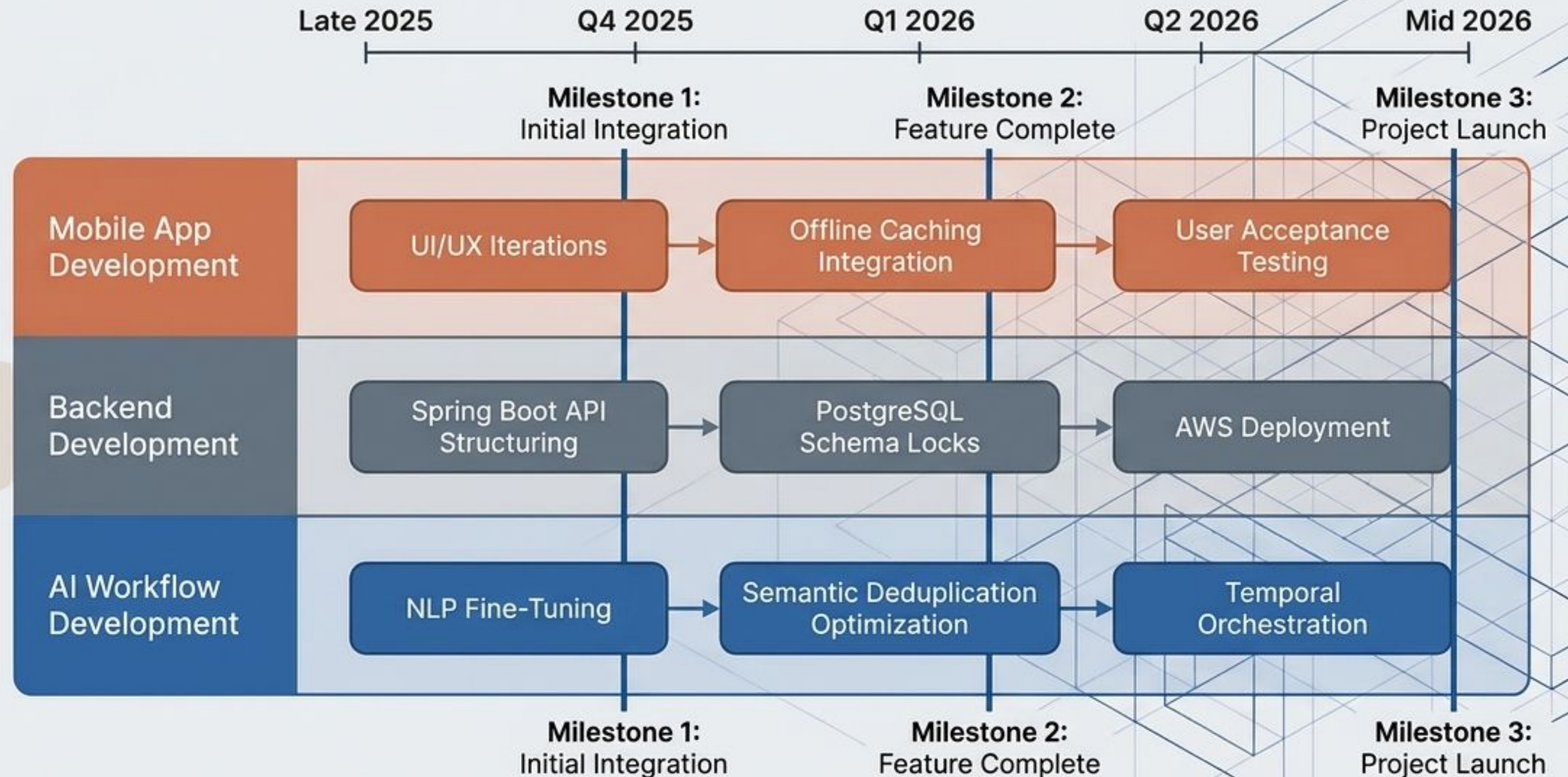
CI/CD pipelines via GitHub Actions, AWS infra scaling, and testing environments.



Project Timeline

Unified development lifecycle spanning **Late 2025** to **Mid 2026**.

Chaos/Real World



Chaos/Real World

Screenshot of Meetings & comments(online/offline) by Guide

WEEK 18	Frontend started Refactoring and built a base RAG AI.	Shi/✓ 20/1/26
WEEK 19	Backend is doing migration of database. AI RAG system can now retrieve policy and understand them.	Shi/✓ 24/1/26
WEEK 20	Backend is doing DDO refactoring. AI optimized RAG infrastructure and progressed with compiler architecture planning. Frontend doing optimization.	Shi/✓ 30/1/26
WEEK 21	Frontend doing bug fixing and building testing suite. Backend doing AWS integration.	Shi/✓ 6/2/26
WEEK 22	Backend focused on load testing, rate limiting and optimizing caching. Frontend worked on dynamic categories and UX updates. AI initiated temporal workflow.	Shi/✓ 11/2/26
WEEK 23	Frontend added offline support functionality. Backend finished media handling logic.	Shi/✓ 17/2/26
WEEK 24	AI finished finalized CRM and end-to-end workflow logic. The initial Beta release was successfully deployed.	Shi/✓ 25/2/26
WEEK 25	Backend completed Dockerization & CI/CD pipelines. AI integration was officially marked as complete.	Shi/✓ 2/3/26
WEEK 26	Full system integration was executed. The team engaged in rigorous QA, bug fixing & end-to-end testing.	Shi/✓ 15/3/26

WEEK 27	Release App version 1.3.0 of Mudda.	Shi/✓ 25/3/26
WEEK 28	Implemented RWA features in Flutter and planned backend integration.	Shi/✓ 1/4/26
WEEK 29	Made token optimized - temporal compatible compiler in AI planning microservices.	Shi/✓ 8/4/26
WEEK 30	Implemented and tested RWA features in FactAPI.	Shi/✓ 13/4/26
WEEK 31	Created token usage dashboard and RAG microservice phantom environments.	Shi/✓ 23/4/26
WEEK 32	Evaluated auto issue categorization pipelines with fine tuned Roberta models.	Shi/✓ 30/4/26
WEEK 33	Refactored web dashboard UI to have a user friendly chat interface.	Shi/✓ 6/5/26
WEEK 34	Development of terraform scripts for easy deployment and migration.	Shi/✓ 11/5/26
WEEK 35	Tested government faced backend with AI, Flutter, NextJS and Spring microservices using Kafka.	Shi/✓ 18/5/26